

Computer Networks(CS3001) Sessional-I Exam

Date: February 25th 2025

Course Instructor(s)

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Total Time (Hrs): 1

Total Marks: 15

Total Questions: 3

Roll No

Section

Student Signature

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Attempt all the questions.

CLO # 2: Apply network protocol and communication services for client/server and other application layouts

Q1: [2 + 2 + 1 = 5 Marks]

- Consider an IoT-based healthcare system where patient vital signs are sent to a cloud server for analysis.
 - Identify which TCP/IP layer(s) is/are most critical for real-time data transmission and justify why.
 - Briefly explain how encapsulation ensures security in this scenario.
- A circuit-switched network requires a setup time of 450 ms and operates at 15 Mbps. A packet-switched network has no setup time but has a per-packet processing delay of 12 ms per packet. A file of 5 MB is being sent, with each packet size of 1 MB. Compute the total transmission time for both networks.
- A user queries www.example.com, and the response takes a total of 300msec, because root, TLD, and authoritative DNS servers are involved. Compute the time saved if the TLD server's response is cached locally.

Q2: [2 + 2 + 1 = 5 Marks]

- A browser loads a webpage with 10 objects from the same server.
 - Compute the total time needed if using non-persistent HTTP with 2 parallel TCP connections and an RTT of 100 ms.
 - How does persistent HTTP with pipelining improve performance?
- A web server handles 3600 requests per hour, with each requested object averaging 500Kb in size. When a caching proxy is introduced, as a result, the hit ratio improves to 80%. Compute the bandwidth saved per hour due to caching.
- In addition to HOL blocking mitigation, and priority based transmission, what is the main factor that resulted in HTTP/2 to provide better efficiency than HTTP1.1.

National University of Computer and Emerging Sciences

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CLO # 1: Describe and evaluate the protocols, services and functions provided by each layer in the Internet protocol stack

Q3:

[1 + 2 + 2 = 5 Marks]

- a) What is the function of the twisting in twisted-pair cable?
- b) Consider the scenario shown in Figure. 1, in which transmission rates and propagation delay at each link are shown. Assume there is no cache, there is no queuing delay at the routers, and the packet processing delays at routers and nodes are all zero. All packets in the network are 10,000 bits long.

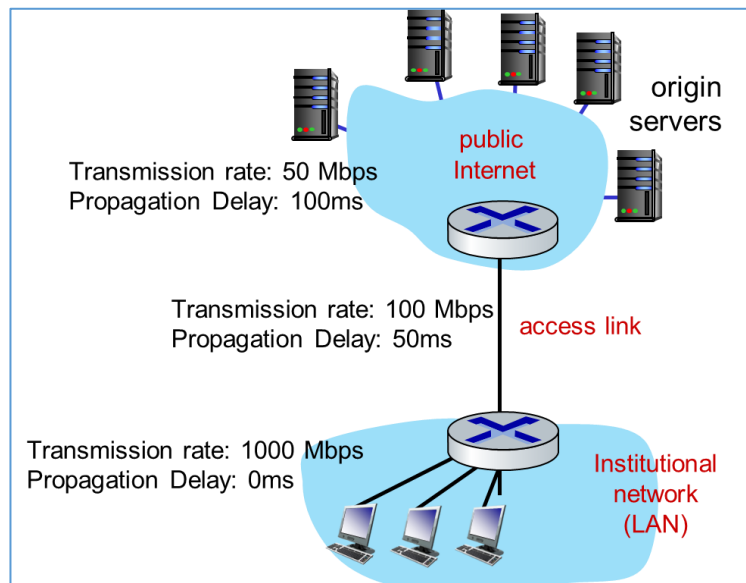


Figure 1. A network scenario

- i. What is the end-to-end delay from when a packet is transmitted by the server to when it is received by the client?
- ii. What is the maximum rate at which the server can deliver data to a single client if we assume no other clients are making requests?
- c) Redraw the below Table in your answer script, and complete it with appropriate entries.

Layer Name	PDU	Device	Protocols	Addressing
Application				
Transport				
Network				
Data Link				

===== Thank you =====